

Joe Chen

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Availability: January - August 2027

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 3.70/4.0 | Sept 2024 - May 2028

Relevant Coursework: Algorithms and Data, Computer Systems, Robotics, Machine Learning and Data Mining 1 & 2, Practical Neural Networks, Probability and Statistics, Linear Algebra

SKILLS

Languages: Python, Java, TypeScript, JavaScript, C, HTML/CSS, SQL

Frameworks & Libraries: PyTorch, NumPy, TensorFlow, React, Node.js, Next.js, FastAPI, UNIX/Linux, Prisma, OpenCV

Tools: Git/GitHub, AWS, Jupyter, Neon, SLURM

EXPERIENCE

Amazon

Boston, MA

Software Development Engineer Intern

June 2026 - Aug 2026

- Decoupled context acquisition from a voice assistant's mic-control request path, running it eagerly in the window between speech recognition and the agent's response, removing the synchronous fetch path entirely so every signal has one source, cutting what the request waits for by ~99%, with missing context degrading the decision rather than failing it
- Designed and built a Java library backing the context cache on AWS ElastiCache for Valkey, provisioned in TypeScript via AWS CDK with IAM authentication, using library-owned models so consumers adopt it without inheriting upstream SDK dependencies
- Engineered an event-driven Java producer that fans out three upstream context fetches as independent tasks on an isolated bounded executor, writing each result the moment it lands so one slow or failed dependency never blocks the other two

Northeastern University

Boston, MA

Teaching Assistant, DS4400: Machine Learning and Data Mining I

Sept 2026 - Dec 2026

Visual Intelligence Lab @ Northeastern University

Boston, MA

Undergraduate Research Assistant

March 2026 - Present

- Developed a synthetic relighting data pipeline in Blender and Infinigen, rendering 3D scenes under environment maps, point lights, and area lights with programmatically varied wattage and color temperature to produce paired training data for relighting research
- Scripted SLERP-interpolated camera trajectories for multi-view capture and deployed the pipeline on a SLURM cluster across A5000 and A6000 GPUs, resolving undocumented dependency conflicts to parallelize rendering

Physical AI Team @ Northeastern AI Club

Boston, MA

Member of Technical Staff

Sept 2026 - Present

PROJECTS

AI Local Assistant | Python, PyTorch, OpenCV, MediaPipe, Faster-Whisper

Sept 2025 - Present

- Quantized Llama-3.2-11B-Vision-Instruct with NF4 double quantization to cut VRAM over 50%, fitting a multimodal LLM, real-time speech recognition, and computer vision on a single consumer GPU
- Trained a Siamese convolutional neural network (CNN) to 98.8% face-verification accuracy using contrastive loss on personal photos plus the LFW dataset, enabling new-user enrollment from a few reference images with no retraining
- Integrated Faster-Whisper speech recognition, OpenCV and MediaPipe vision, and LLM inference into a thread-safe pipeline with non-blocking queues, keeping interaction real-time with zero cloud dependency

Resona - Music Analytics App | TypeScript, Next.js, PostgreSQL, Prisma, Python, FastAPI

Sept 2025 - Present

- Shipped a full-stack Next.js and PostgreSQL app where users rate tracks, albums, and artists 0-10, with Google OAuth via NextAuth v5 and a custom Prisma schema modeling ratings, threaded comments, likes, and a follower graph
- Implemented a FastAPI microservice training Funk SVD matrix factorization on GPU in PyTorch, producing latent user vectors for recommendation and taste-similarity matching, with on-demand and nightly retraining